MULTINOMIAL THEOREM

The Multinomial Theorem

The binomial theorem for a base with r summands.

THEOREM

$(x_1 + \cdots + x_r)^n = \sum \binom{n}{k_1, \ldots, k_r} x_1^{k_1} \cdots x_r^{k_r}$, the sum over all non-negative $k_1 + \cdots + k_r = n$.

1 **THE COEFFICIENT**

$\binom{n}{k_1, \ldots, k_r} = \dfrac{n!}{k_1! \cdots k_r!}$: the ways to partition n items into r labelled groups of the given sizes.

2 **BINOMIAL CASE**

$r = 2$: $\binom{n}{k,\, n-k}$ collapses to $\binom{n}{k}$.

3 **NUMBER OF TERMS**

$\binom{n+r-1}{r-1}$, the number of weak compositions of n into r parts; for $r = 2$ this is $n + 1$.

4 **EXAMPLE**

$(a + b + c)^3$: $r = 3$, so $\binom{5}{2} = 10$ distinct terms.

COUNTING $n + 1$ TERMS FOR THREE SUMMANDS

Counting the terms of $(a + b + c)^3$ as $3 + 1 = 4$ carries the binomial count over to a base with three summands. With $r = 3$ the count is $\binom{3+2}{2} = \binom{5}{2} = 10$.